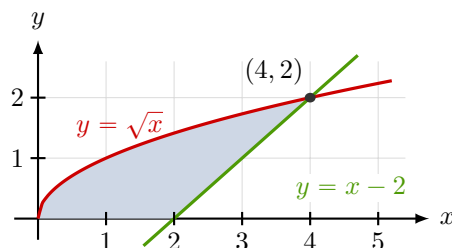


L4.1c Areas Along y

Work in your group. Check your answers against the Quarto tonight.

The sideways rectangle

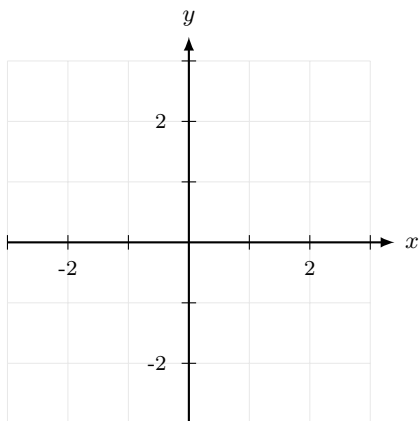
1. The region bounded by $y = \sqrt{x}$, $y = x - 2$ and the x -axis.



- (a) Solve each boundary for x .
- (b) Draw **one** sideways rectangle on the figure. Test a value of y to decide which boundary is x_R and which is x_L .
- (c) Write **one** integral for the area and evaluate it.
- (d) Set up the same area with upright rectangles. How many integrals does it take? Do not evaluate.

2. Find the area of the region enclosed by $x = y^2 - 2$ and $x = y$.

3. Sketch $x = 1 - y^2$ and $x = y^2 - 1$ on the grid, then find the area of the region they enclose.



Which rectangle, and where to split

4. For each region, draw one upright and one sideways rectangle. Decide which kind crosses the region in one piece, and write that integral. Do not evaluate.

(a) The region enclosed by $y = x^2$ and $y = x + 2$.

(b) The region bounded by $y = \sqrt{x}$, $y = 2 - x$ and the x -axis.

5. Two regions, one question.

(a) The region between $x = 4 - y^2$ and the y -axis. Sketch it and find its area.

(b) The region between $x = \cos y$ and the y -axis for $0 \leq y \leq \pi$. Sketch it and find its area.

(c) In (a) the boundary turns around inside the interval and you wrote **one** integral. In (b) you had to **split**. In one sentence, what is the difference between the two situations?